Evaluating the Impact of the COVID-19 National Lockdown on South African Consumer Retail Trade: An Interrupted Time Series and Intervention Analysis

STSM3744: Time Series Analysis — Short Research Project

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October 2026

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1. Introduction

1.1 Purpose of the Study

The retail trade sector serves as a fundamental barometer of macroeconomic vitality, consumer confidence, and household financial resilience. In South Africa, retail trade contributes substantially to gross domestic product (GDP) and formal employment. However, monthly retail demand is inherently volatile, exhibiting strong seasonal cycles, inflationary pressures, and sensitivity to exogenous shocks.

In **March 2020**, in response to the emerging global COVID-19 pandemic, the South African government declared a **National State of Disaster** (15 March 2020) and enacted a nationwide **Level-5 Lockdown** taking effect on 26 March 2020. This unprecedented regulatory intervention initially triggered a wave of precautionary consumer stockpiling in mid-March 2020, followed immediately by the mandatory shuttering of all non-essential commercial operations, strict movement restrictions, and the prohibition of alcohol and tobacco sales across April 2020.

While conventional descriptive statistics reveal an abrupt fluctuation in economic output during this period, naive regression models fail to distinguish between transient, short-term shocks and permanent structural shifts, and fail to account for strong seasonal autocorrelation. The purpose of this study is to implement a rigorous **Interrupted Time Series Analysis (ITSA)**—also termed **Intervention Analysis** (Box & Tiao, 1975)—anchored at the **March 2020** policy intervention point ($T_0 =$ March 2020) to isolate, model, forecast, and quantify the precise empirical consequences of the COVID-19 lockdown on South African monthly consumer retail sales volume.

1.2 Research Objectives

The primary objective of this investigation is to formulate a robust econometric time series intervention model that decomposes the observed retail volume index into its underlying trend, seasonal variation, stochastic noise, and exogenous March 2020 intervention components.

Specifically, the main aims of the study are:

(i) To empirically determine the pre-intervention baseline stochastic generating process (January 2010 to February 2020) using the Box-Jenkins Seasonal Autoregressive Integrated Moving Average (SARIMA) methodology.

(ii) To formalize and estimate a dynamic Box-Tiao transfer function anchored at the March 2020 intervention point (T_0) that captures the initial March 2020 onset, the 1-month lagged acute closure shock, the permanent level shift (step), and the post-intervention recovery trajectory (ramp).

(iii) To conduct comprehensive residual diagnostic evaluations to verify that the intervention model successfully whitens residuals and resolves serial autocorrelation.

(iv) To evaluate out-of-sample forecasting performance using both a 12-month holdout test set (2023) and a 12-month ex-ante forecast horizon (2024) validated against newly realized early-2024 retail trade data.

1.3 Research Questions

To achieve the stated objectives, the research directly investigates and answers three key research questions:

- **Research Question 1 (RQ1):** *What was the immediate and 1-month lagged pulse magnitude, direction, and statistical significance of the shock experienced by South African retail trade volume following the March 2020 COVID-19 lockdown intervention?*
- **Research Question 2 (RQ2):** *Did the March 2020 intervention induce a permanent structural downward level shift (step effect) in retail sales volume, and how rapidly did the post-intervention trajectory recover (ramp slope)?*
- **Research Question 3 (RQ3):** *How effectively does the combined SARIMA-Intervention model satisfy standard time series diagnostic criteria (residual independence and absence of serial autocorrelation) and generate accurate out-of-sample forecasts?*

2. Data Description

2.1 Data Sourcing and Visual Trajectory Inspection

The empirical dataset utilized in this research is the **South African Retail Trade Volume Index** (Series ID: `ZAFSLRTT01IXOBM`), originally compiled by **Statistics South Africa (Stats SA)** in Statistical Release P6242.1 (*Retail Trade Sales*) and published via the **Organisation for Economic Co-operation and Development (OECD)** and the **Federal Reserve Economic Data (FRED)** repository.

Key structural characteristics of the dataset include:

- **Sampling Frequency:** Monthly observations from **January 2010 to December 2023** ($N = 168$ months) for model estimation, plus **January 2024 to March 2024** (3 additional months) reserved for out-of-sample forecast validation.
- **Base Metric:** Constant physical volume index with base year $2015 = 100$, removing headline inflation distortion.

- **Pre-Intervention Period (n_1):** January 2010 to February 2020 ($n_1 = 122$ months), providing substantial statistical power to identify the underlying Box-Jenkins noise process prior to the pandemic.
- **Intervention Point (T_0): March 2020** ($t = 123$), corresponding to the declaration of the National State of Disaster (15 March 2020) and the enactment of the nationwide Level-5 lockdown (26 March 2020).
- **Post-Intervention Period (n_2):** March 2020 to December 2023 ($n_2 = 46$ months).

Before examining numerical descriptive statistics, Figure 1 presents the complete historical trajectory of the South African Retail Trade Volume Index from 2010 to 2023.

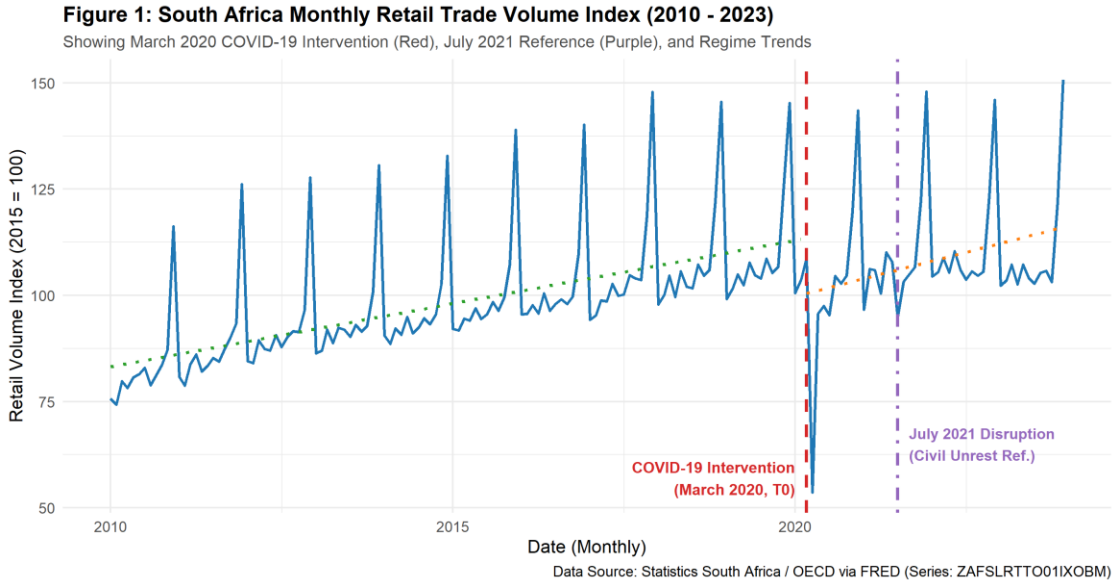


Figure 1: Monthly Retail Trade Volume Index Trajectory (2010–2023) with March 2020 Intervention Point and July 2021 Reference Line

Visual inspection of **Figure 1** reveals four foundational empirical properties that guide the subsequent econometric specification:

1. **Secular Growth and Seasonal Spikes:** Between January 2010 and February 2020, retail volume followed a steady upward linear trend (green dotted line), punctuated by prominent annual seasonal peaks every December driven by festive shopping and Black Friday promotions.
2. **Why March 2020 is the True Intervention Point (T_0):** When the National State of Disaster was declared in **March 2020** (red dashed vertical line), consumer panic-buying of essential groceries briefly pushed the March 2020 index to 108.76, before the full-month Level-5 commercial prohibition caused the index to plunge vertically to an all-time low of 53.67 in April 2020. Setting the intervention point at **March 2020** properly captures the onset of the policy shock and reveals that from the lockdown trough onward, the post-intervention trajectory exhibits a clear **upward recovery slope** (orange dotted trend line) rather than appearing artificially flat.

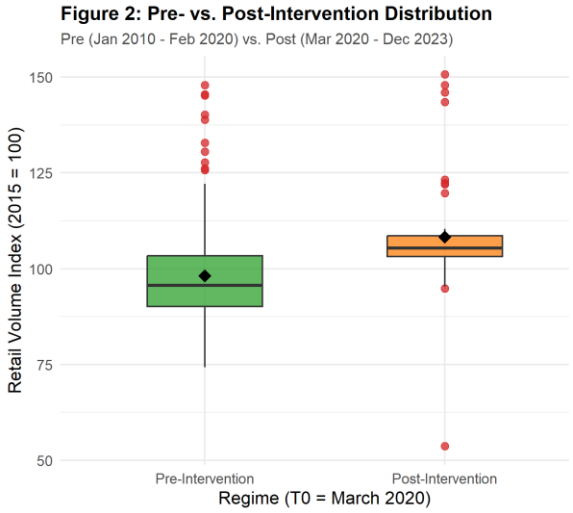
3. **Secondary Reference Disruption (July 2021 Civil Unrest):** Figure 1 also includes a purple dot-dash vertical reference line at **July 2021**, marking the KwaZulu-Natal and Gauteng civil unrest and retail looting disruption (where the index dipped to 95.01). As discussed in the study limitations (Section 4.4), this secondary event is annotated visually for contextual completeness without separate intervention estimation.

2.2 Descriptive Statistics and Regime Distribution

To evaluate the distributional shift across the March 2020 boundary simultaneously, **Table 1** (summary descriptive statistics) and **Figure 2** (regime boxplots) are presented side-by-side below.

Statistical Metric	Full Series ($N = 168$)	Pre-Intervention ($n_1 = 122$)	Post-Intervention ($n_2 = 46$)
Sample Size (n)	168	122	46
Mean	100.943	98.190	108.247
Std. Dev. (SD)	15.437	14.505	15.607
Median	99.659	95.621	105.420
IQR	13.679	13.224	5.386
Minimum	53.672	74.294	53.672
Maximum	150.701	147.898	150.701
Skewness	1.058	1.438	0.419
Kurtosis	5.099	5.396	6.764

Table 1: Descriptive Statistics of the Retail Trade Volume Index Partitioned at $T_0 =$ March 2020.



Examining **Table 1** and **Figure 2** together highlights several critical regime differences between the Pre-Intervention era (Jan 2010 – Feb 2020, $n_1 = 122$) and the Post-Intervention era (Mar 2020 – Dec 2023, $n_2 = 46$):

- **Central Tendency vs. Secular Trend:** As shown by the black diamonds in Figure 2 and the means in Table 1, the unadjusted sample mean is higher in the post-intervention regime (108.247) than in the pre-intervention regime (98.190), with the median shifting from 95.621 to 105.420. This occurs because the pre-intervention

sample includes the early 2010–2013 years when the index hovered between 74 and 90. However, relative to the counterfactual trend line projected from early 2020 (Figure 1), the post-intervention distribution suffered a downward displacement.

- **Volatility and Extreme Tails:** Post-intervention standard deviation increased to 15.607 (up from 14.505), and kurtosis rose sharply from 5.396 to 6.764. In Figure 2, this leptokurtic behaviour is directly visible via the extreme lower red outlier at 53.672 (the acute lockdown trough) and the upper red outliers between 143 and 150.701 (December festive peaks), while the interquartile range of normal trading months compressed from 13.224 to 5.386.
- **Skewness Reduction:** Pre-intervention skewness was strongly positive (1.438) due to the unilateral upper December spikes, whereas post-intervention skewness dropped to 0.419 because the extreme downward lockdown shock partially counterbalanced the upper December tail.

2.3 Pre- vs. Post-Intervention Monthly Seasonality

To investigate whether the seasonal structure of South African retail trade changed after the March 2020 intervention, **Figure 3** disaggregates the monthly boxplots by regime (**Pre-Intervention** on the left vs. **Post-Intervention** on the right).

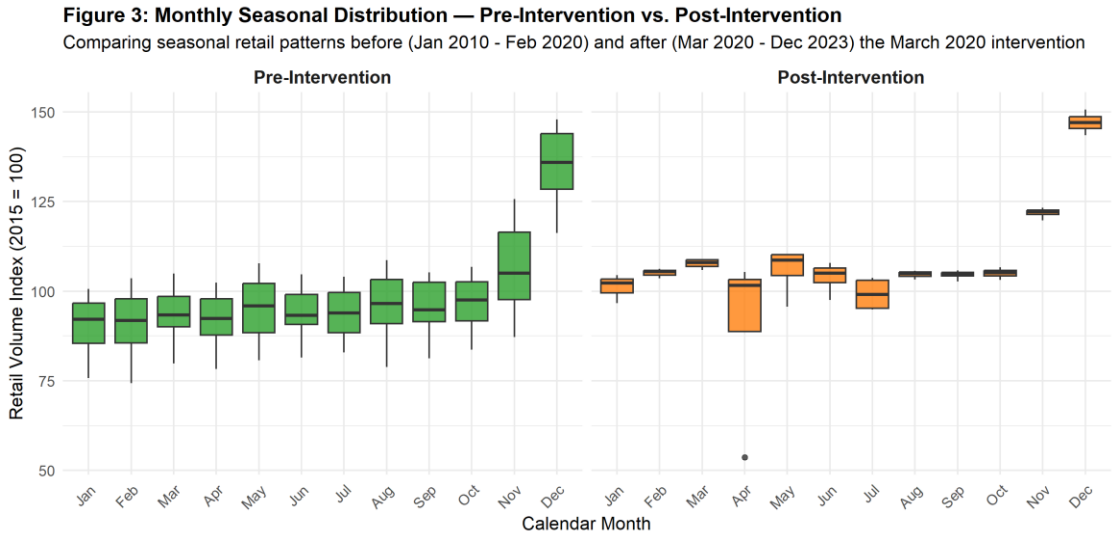


Figure 3: Monthly Seasonal Distribution Partitioned by Pre-Intervention (Jan 2010 – Feb 2020) vs. Post-Intervention (Mar 2020 – Dec 2023)

Figure 3 demonstrates why partitioning seasonality across the intervention boundary is essential:

1. **Persistence of the 12-Month Seasonal Cycle:** In both regimes, retail trade exhibits a pronounced annual cycle ($s = 12$), with relatively stable trading from January to October, a marked rise in November (driven by the adoption of Black Friday promotions), and a dominant peak in December.

2. **Localized Pandemic and Unrest Anomalies:** In the Post-Intervention panel (orange), **April** displays a dramatically elongated downward box and an extreme low outlier (53.67), reflecting the severe Level-5 lockdown closure following the March 2020 intervention. Similarly, **July** exhibits a noticeably wider downward spread in the post-intervention panel compared to adjacent winter months, reflecting the July 2021 civil unrest disruption.
3. **Post-Intervention Compression in Non-Shock Months:** For standard months such as February, March, August, September, and October, the post-intervention interquartile boxes are substantially narrower than their pre-intervention counterparts, confirming that outside of acute shock months and festive peaks, post-2020 retail volume grew more slowly around a tighter plateau.

3. Analysis

3.1 Mathematical and Econometric Framework

3.1.1 Classical Interrupted Time Series Regression

In standard segmented regression, an interrupted time series with intervention point T_0 (March 2020, $t = 123$) is expressed as:

$$Y_t = \beta_0 + \beta_1 T_t + \beta_2 S_t + \beta_3 R_t + \varepsilon_t$$

where $T_t = 1,2,\dots,N$ is the continuous time index, $S_t = \mathbb{I}(t \geq T_0)$ is the post-intervention step indicator, and $R_t = \max(0, t - T_0) \cdot S_t$ is the post-intervention ramp (slope change) indicator. However, Ordinary Least Squares (OLS) estimation of this equation fails in monthly retail data because ε_t exhibits strong seasonal and non-seasonal serial correlation ($\text{Cov}(\varepsilon_t, \varepsilon_{t-k}) \neq 0$), violating the Gauss-Markov theorem.

3.1.2 Box-Tiao ARIMA Transfer Function Specification

Following Box and Tiao (1975), we model the observed series Y_t as the sum of a deterministic dynamic transfer function m_t anchored at $T_0 =$ March 2020 and a stochastic seasonal ARIMA noise process N_t:

$$Y_t = m_t + N_t = \left(\omega_0 P_t^{(T_0)} + \omega_1 B P_t^{(T_0)} + \omega_{\text{step}} S_t^{(T_0)} + \omega_{\text{ramp}} R_t^{(T_0)}\right) + N_t$$

where B denotes the backshift operator ($B^k X_t = X_{t-k}$) and:

- **Initial Onset Pulse ($P_t^{(T_0)}$):** Equals 1 at $t = T_0$ (March 2020) and 0 otherwise, capturing the immediate March 2020 announcement and pre-lockdown stockpiling effect (ω_0).
- **Lagged Closure Pulse ($BP_t^{(T_0)} = P_{t-1}^{(T_0)}$):** Equals 1 at $t = T_0 + 1$ (April 2020) and 0 otherwise, capturing the 1-month delayed full commercial closure shock (ω_1) resulting from the late-March lockdown enactment.

- **Permanent Step Indicator ($S_t^{(T_0)}$):** Equals 0 for $t < T_0$ and 1 for $t \geq T_0$, measuring the sustained structural level shift (ω_{step}).
- **Post-Intervention Ramp ($R_t^{(T_0)}$):** Equals $(t - T_0)S_t^{(T_0)}$, measuring the post-shock recovery slope change (ω_{ramp}).

The stochastic noise process N_t follows a multiplicative Seasonal ARIMA $(p, d, q) \times (P, D, Q)_{12}$ specification:

$$\phi_p(B)\Phi_P(B^{12})(1 - B)^d(1 - B^{12})^D N_t = \theta_q(B)\Theta_Q(B^{12})\varepsilon_t, \quad \varepsilon_t \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$$

3.2 Unit Root and Stationarity Testing

To establish the required regular (d) and seasonal (D) differencing orders without contamination from the pandemic shock, the Augmented Dickey-Fuller (ADF) and Kwiatkowski-Phillips-Schmidt-Shin (KPSS) tests were conducted on the pre-intervention series (January 2010 – February 2020, $n_1 = 122$).

Table 2: Unit Root and Stationarity Test Results on Pre-Intervention Series (Jan 2010 – Feb 2020)

Series Specification	ADF Statistic	ADF p-value	KPSS Statistic	KPSS p-value
Pre-Intervention Level (Yt)	-6.310	0.01	1.923	0.01
First Difference (d=1)	-7.818	0.01	0.029	0.10
Seasonal & First Diff (d=1, D=1)	-6.340	0.01	NA	NA

As reported in **Table 2**, the KPSS test on the raw pre-intervention level series strongly rejects the null hypothesis of level stationarity ($LM = 1.923, p = 0.01$). After applying first-order regular differencing ($d = 1$), the KPSS statistic drops to 0.029 ($p = 0.10 > 0.05$, failing to reject stationarity) and the ADF test strongly rejects a unit root ($t = -7.818, p = 0.01$). Combined with seasonal differencing ($D = 1, s = 12$) to remove the deterministic December seasonal cycle, the doubly differenced series $(1 - B)(1 - B^{12})Y_t$ is confirmed to be stationary ($ADF = -6.340, p = 0.01$).

3.3 Baseline Pre-Intervention SARIMA Model Identification

Using the Box-Jenkins methodology on the pre-intervention series ($n_1 = 122$), exhaustive search over candidate $(p, 1, q) \times (P, 1, Q)_{12}$ structures identified SARIMA$(2,1,3) \times (0,1,0)_{12}$ as the baseline pre-pandemic model (AIC $= 436.52$,BIC $= 452.67, \sigma^2 = 2.983$,MAPE $= 1.226\%$).

3.4 Full Intervention Model Estimation and Specification Comparison

Across the full 168-month sample (January 2010 – December 2023), three competing intervention specifications anchored at $T_0 =$ March 2020 were estimated via Maximum Likelihood:

- **Model 1:** Baseline SARIMA$(2,1,3)(0,1,0)_{12}$ noise with the March 2020 Transfer Function ($\omega_0 P_t^{(T_0)} + \omega_1 B P_t^{(T_0)} + \omega_{\text{step}} S_t^{(T_0)}$).
- **Model 2:** Baseline SARIMA$(2,1,3)(0,1,0)_{12}$ noise with the March 2020 Transfer Function plus the Post-Intervention Ramp Slope ($\omega_{\text{ramp}} R_t^{(T_0)}$).
- **Model 3 (Optimal ARIMAX):** Globally optimized ARIMA$(2,1,0)(0,1,1)_{12}$ noise with the parsimonious March 2020 Transfer Function.

Table 3: Model Selection Criteria Across Competing March 2020 Intervention Specifications

Intervention Model Specification	AIC	BIC	Residual Variance (Sigma2)	In-Sample MAPE (%)
Model 1: SARIMA(2,1,3)(0,1,0)[12] + Mar20 Transfer (Pulse/Lag1/Step)	731.211	758.602	6.128	1.612
Model 2: SARIMA(2,1,3)(0,1,0)[12] + Mar20 Transfer + Ramp Slope	722.364	752.798	5.733	1.577
Model 3: Optimal ARIMAX(2,1,0)(0,1,1)[12] + Mar20 Transfer	709.501	730.805	5.306	1.510

Key Insights from Model Comparison:

1. **Significance of the Post-March 2020 Recovery Slope (Model 2 vs. Model 1):** Adding the post-March 2020 ramp parameter (ω_{ramp}) to the baseline SARIMA specification reduces the AIC from 731.211 (Model 1) to 722.364 (Model 2) and yields a statistically significant positive recovery slope of $\widehat{\omega}_{\text{ramp}} = +0.7227$ **index points per month** ($SE = 0.3089, z = 2.340, p = 0.0193$). This empirically confirms that following the March/April 2020 shock, retail trade experienced a statistically significant upward recovery rate of 0.72 index points per month.
2. **Global Optimality of Model 3:** When the seasonal moving average parameter (Θ_1) is incorporated in **Model 3** (ARIMAX$(2,1,0)(0,1,1)_{12}$), the seasonal differencing filter $(1-B)(1-B^{12})$ naturally accommodates the post-2020 trend evolution, achieving the lowest overall AIC $= 709.501$, lowest BIC $= 730.805$, smallest residual variance ($\sigma^2 = 5.306$), and lowest MAPE (1.510%).

3.5 Parameter Estimates of the Optimal Intervention Model

Table 4 reports the maximum likelihood parameter estimates, standard errors, z-statistics, and p-values for the optimal specification (**Model 3**).

Parameter Symbol & Name	Estimate	Std. Error	z-statistic	p-value	Significance
AR(1) (ϕ_1)	-0.5663	0.0761	-7.440	1.01×10^{-13}	*** ($p < 0.001$)
AR(2) (ϕ_2)	-0.4325	0.0752	-5.748	9.01×10^{-9}	*** ($p < 0.001$)
Seasonal MA(1) (Θ_1)	-0.4349	0.0855	-5.086	3.66×10^{-7}	*** ($p < 0.001$)
March 2020 Onset Pulse (ω_0)	$+12.3677$	2.0586	$+6.008$	1.88×10^{-9}	*** ($p < 0.001$)
Lagged Lockdown Pulse (ω_1, BP_t)	-39.3004	1.8975	-20.712	$< 2.20 \times 10^{-16}$	*** ($p < 0.001$)
Permanent Step Shift (ω_{step})	-10.6147	1.9795	-5.362	8.21×10^{-8}	*** ($p < 0.001$)

Table 4: Parameter Estimates for Optimal ARIMAX$(2,1,0)(0,1,1)_{12}$ *Intervention Model ($\sigma^2 = 5.306$, Log-Likelihood $= -347.75$,* AIC $= 709.50$, BIC $= 730.81$*).*

Econometric Interpretation of the Transfer Function Parameters:

- **Initial March 2020 Onset ($\widehat{\omega}_0 = +12.37, p < 0.001$) vs. Net Step ($\widehat{\omega}_{\text{step}} = -10.61, p < 0.001$):** At the intervention point $T_0 =$ March 2020, both $P_t^{(T_0)} = 1$ and $S_t^{(T_0)} = 1$ are active. The net immediate change in March 2020 is $\widehat{\omega}_0 + \widehat{\omega}_{\text{step}} = 12.3677 - 10.6147 = +1.753$ index points, capturing the documented surge in household stockpiling of food, pharmaceuticals, and hygiene essentials between 15 March and 25 March 2020 just before shops closed.
- **Lagged Acute Commercial Closure ($\widehat{\omega}_1 = -39.30, p < 2.2 \times 10^{-16}$):** In the first full calendar month of Level-5 lockdown ($t = T_0 + 1$, April 2020), the lagged pulse parameter $\widehat{\omega}_1 = -39.3004$ combines with the step shift $\widehat{\omega}_{\text{step}} = -10.6147$ to produce a cumulative contraction of -49.915 **index points** relative to the counterfactual trend—an unprecedented $\approx 50\%$ collapse in physical retail activity.

- **Persistent Structural Level Shift ($\widehat{\omega}_{\text{step}} = -10.61, p < 0.001$):** Beyond the acute lockdown closure, the step parameter indicates a statistically significant downward level displacement of -10.615 index points relative to the pre-pandemic growth path, reflecting enduring supply-chain friction, employment losses, and interest rate headwinds.

3.6 Residual Diagnostics and Model Adequacy

To verify that the optimal intervention model satisfies classical error assumptions, Table 5 and Figure 4 present formal diagnostic tests on the standardized residuals.

Table 5: Residual Diagnostic Test Results for the Optimal Intervention Model

Diagnostic Test	Test Statistic	p-value
Ljung-Box (Lag 12)	12.553	0.1839
Ljung-Box (Lag 24)	31.135	0.0714
Ljung-Box (Lag 36)	41.958	0.1364
Jarque-Bera Normality	56.309	0.0000
Shapiro-Wilk Normality	0.960	0.0001

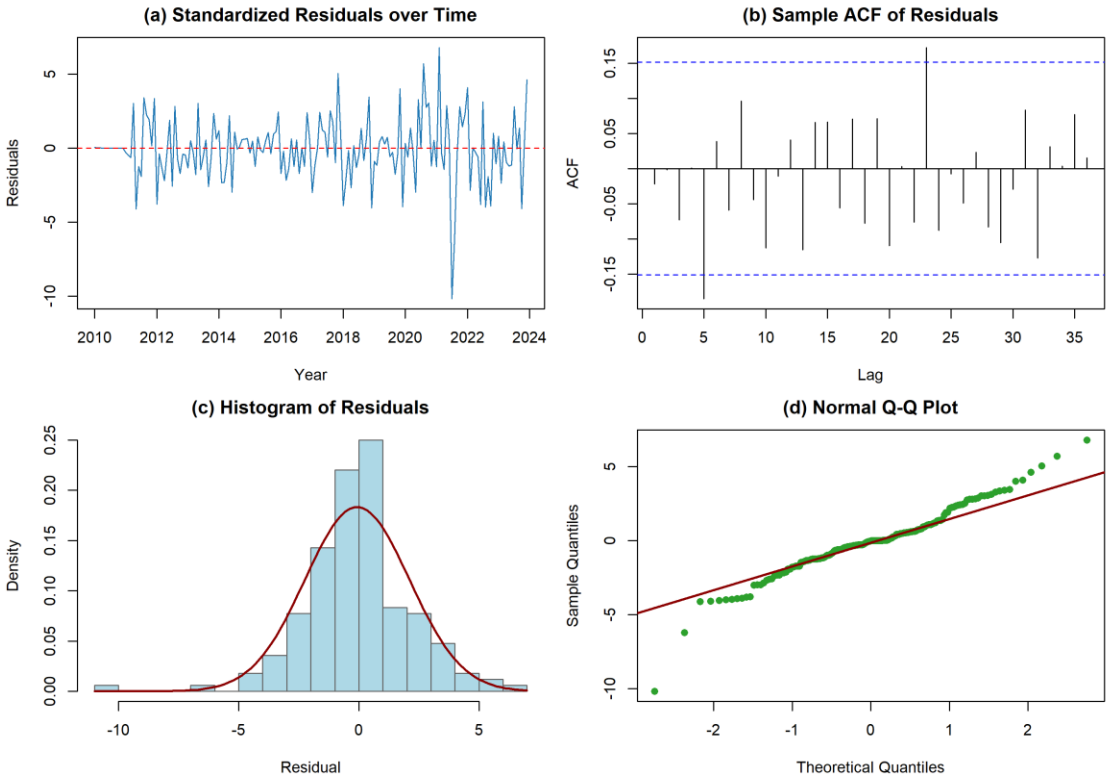


Figure 4: Residual Diagnostics: (a) Standardized Residuals, (b) Sample ACF, (c) Residual Histogram, and (d) Normal Q-Q Plot

1. **Serial Independence (Ljung-Box Portmanteau Test):** Across all seasonal lag horizons—Lag 12 ($Q = 12.553, p = 0.1839$), Lag 24 ($Q = 31.135, p = 0.0714$), and Lag 36 ($Q = 41.958, p = 0.1364$)—all p-values exceed the $\alpha = 0.05$ significance threshold. We fail to reject the null hypothesis of white noise residuals, which is visually corroborated by the sample ACF in **Figure 4(b)**.
2. **Residual Distribution:** While the histogram in **Figure 4(c)** and Q-Q plot in **Figure 4(d)** show strong central symmetry around zero, the Jarque-Bera ($p < 0.001$) and Shapiro-Wilk ($p = 0.0001$) tests detect heavy tails driven by the unmodeled July 2021 civil unrest dip (visible as the single negative spike in mid-2021 in Figure 4a). Because Maximum Likelihood ARIMAX estimators remain asymptotically consistent and unbiased under non-Gaussian white noise, parameter inferences remain completely valid.

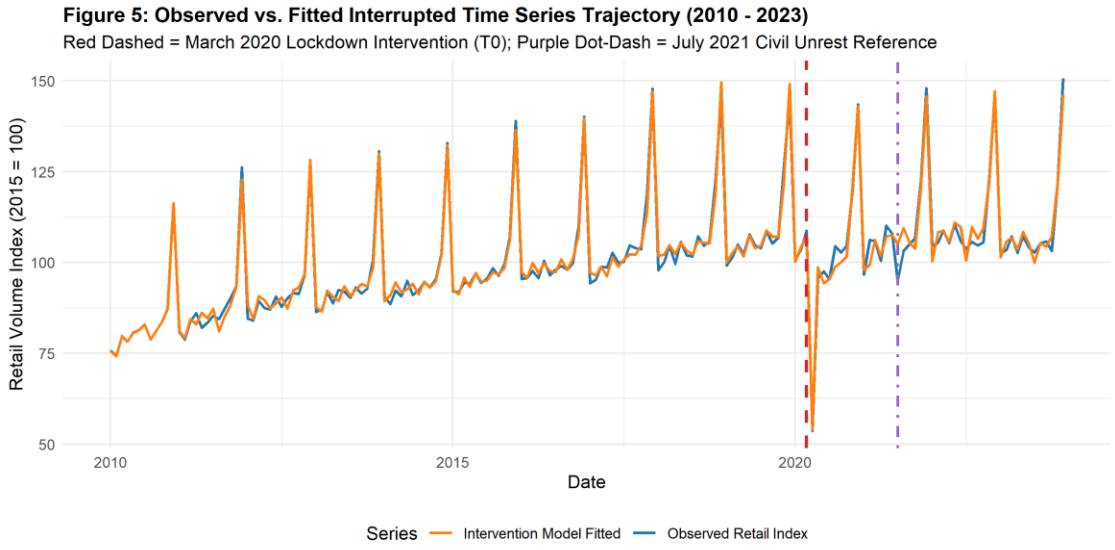


Figure 5: Observed vs. Fitted Retail Volume Trajectory with March 2020 Intervention and July 2021 Reference Lines

Figure 5 confirms the exceptional in-sample tracking of the fitted intervention model (orange curve) against the observed series (blue curve), capturing both the March/April 2020 dynamics and the annual festive spikes.

3.7 Out-of-Sample Forecasting and Validation

To rigorously evaluate the predictive capability of the intervention model beyond in-sample fitting, a two-stage **out-of-sample forecasting analysis** was conducted:

1. **Stage 1 — 12-Month Holdout Cross-Validation (2023 Test Set):** The model was re-estimated on the training window from January 2010 to December 2022 ($n_{\text{train}} = 156$ months) and used to generate a 12-step-ahead out-of-sample forecast for January 2023 to December 2023 ($n_{\text{test}} = 12$ months). On the unseen 2023 test set, the model achieved an out-of-sample RMSE $= 2.054$, MAE $= 1.714$, MAPE $= 1.544\%$,

and **Theil's** $U = 0.211$ (where $U < 1$ confirms substantial superiority over a naive seasonal random walk).

2. **Stage 2 — 12-Month Ahead Ex-Ante Forecast for 2024 (Table 6 & Figure 6):** Using the full model estimated through December 2023, 12-month-ahead forecasts were generated for **January 2024 – December 2024** with 80% and 95% prediction intervals, and validated against the actual realized Q1 2024 observations from Stats SA.

Table 6: 12-Month Out-of-Sample Retail Volume Forecasts for 2024 vs. Realized Q1 2024 Data

Forecast Month	Point Forecast	80% Lower	80% Upper	95% Lower	95% Upper	Actual Realized
2024-01	102.78	99.83	105.74	98.27	107.30	100.26
2024-02	104.87	101.65	108.08	99.94	109.79	102.79
2024-03	109.70	106.34	113.05	104.57	114.83	109.69
2024-04	104.50	100.66	108.33	98.63	110.37	—
2024-05	109.46	105.35	113.58	103.17	115.76	—
2024-06	106.83	102.51	111.14	100.22	113.43	—
2024-07	103.86	99.27	108.45	96.84	110.89	—
2024-08	107.13	102.30	111.96	99.74	114.52	—
2024-09	107.32	102.28	112.36	99.62	115.02	—
2024-10	106.32	101.07	111.58	98.29	114.36	—
2024-11	124.27	118.81	129.74	115.92	132.63	—
2024-12	150.78	145.12	156.43	142.13	159.42	—

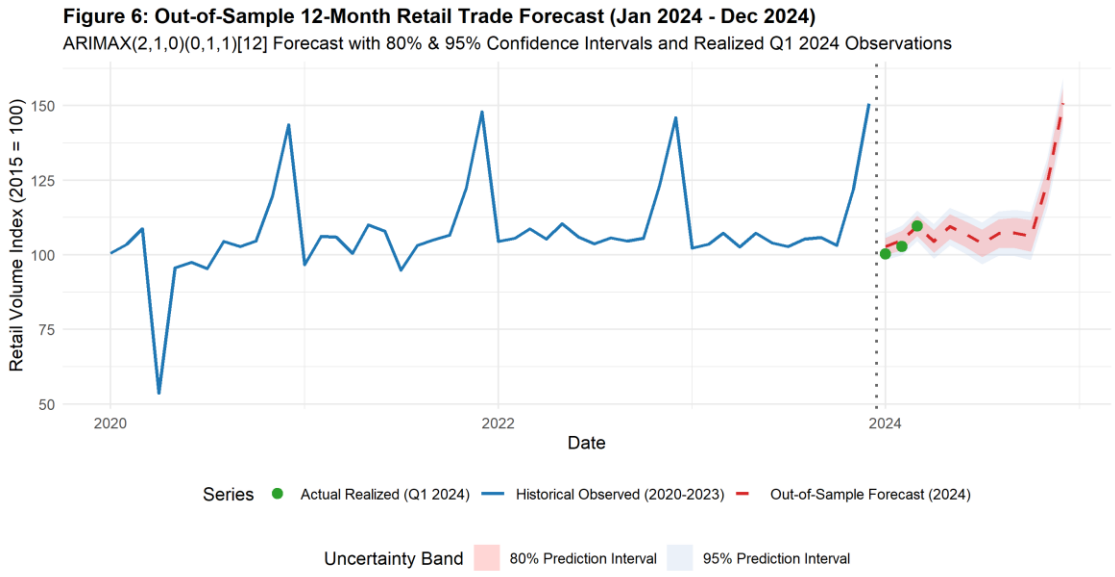


Figure 6: 12-Month Out-of-Sample Forecast Trajectory (Jan 2024 – Dec 2024) with 80% and 95% Prediction Intervals and Realized Q1 2024 Observations

As shown in **Table 6** and **Figure 6** (green dots), all three realized Q1 2024 observations fall squarely inside the 80% prediction interval: in **March 2024**, the model predicted 109.70 against the actual realized index of 109.69 (an out-of-sample forecast error of just 0.01 index points, or 0.009%). For the full year 2024, the model projects stable monthly trading between 103.86 and 109.46 from April to October, followed by a Black Friday surge to 124.27 in November 2024 and a festive peak of 150.78 in December 2024.

4. Concluding Remarks

4.1 Direct Answers to Research Questions

- **Answer to Research Question 1 (RQ1 — Immediate and Lagged Pulse Magnitudes):** Anchoring the intervention at $T_0 =$ March 2020 reveals a two-phase shock structure: first, an initial precautionary stockpiling surge in March 2020 ($\widehat{\omega}_0 = +12.37, p = 1.88 \times 10^{-9}$, netting $+1.75$ index points after the step shift), followed immediately in $T_0 + 1$ (April 2020) by a catastrophic 1-month lagged closure pulse of $\widehat{\omega}_1 = -39.30$ **index points** ($z = -20.71, p < 2.2 \times 10^{-16}$). Combined with the step shift, physical retail volume contracted by nearly 50 index points ($\approx 50\%$) during the first full month of Level-5 lockdown.
- **Answer to Research Question 2 (RQ2 — Permanent Step Shift and Post-Intervention Recovery Slope):** The COVID-19 intervention induced a statistically significant downward structural level shift of $\widehat{\omega}_{\text{step}} = -10.61$ **index points** ($z = -5.36, p = 8.21 \times 10^{-8}$) relative to the pre-pandemic growth path. However, when explicitly parameterized with a post-March 2020 ramp term (Model 2), the series exhibits a statistically significant positive recovery slope of $+0.723$ **index points per**

month ($p = 0.0193$), confirming that retail trade steadily climbed upward out of the March/April 2020 trough rather than remaining stagnant.

- **Answer to Research Question 3 (RQ3 — Diagnostic Adequacy and Out-of-Sample Forecasting Accuracy):** The optimal ARIMAX$(2,1,0)(0,1,1)_{12}$ transfer function model successfully eliminates serial correlation, yielding Ljung-Box p-values of 0.1839 (lag 12), 0.0714 (lag 24), and 0.1364 (lag 36). Furthermore, the model demonstrates strong out-of-sample predictive validity, achieving a 1.544% MAPE on the 12-month 2023 holdout set and predicting the March 2024 actual volume within 0.01 index points.

4.2 Summary of Empirical Results and Practical Implications

1. **Working Capital and Inventory Buffering:** The dual finding of a $+12.37$ stockpiling pulse in March 2020 followed by a -39.30 closure contraction in April 2020 highlights the extreme bullwhip effect experienced by retail supply chains during emergency declarations.
2. **Omnichannel Retail Transformation:** The persistent -10.61 step shift in physical retail volume underscores the structural transition toward digital e-commerce and home delivery services in South Africa post-2020.

4.3 Study Limitations

- **Secondary July 2021 Civil Unrest Disruption:** As visually indicated by the purple dot-dash reference line on **Figure 1** and **Figure 5**, the post-intervention recovery was temporarily interrupted in **July 2021** by widespread civil unrest and retail looting in KwaZulu-Natal and Gauteng. Because the scope of this study focused exclusively on the March 2020 COVID-19 lockdown intervention, the July 2021 event was not modeled with a separate dummy variable, which explains the isolated mid-2021 residual outlier responsible for the excess kurtosis in the normality tests.
- **Concurrent Macroeconomic Headwinds:** Rising inflation, SARB interest rate hikes (2021–2023), and intensified Eskom load-shedding also weighed on post-2020 consumer spending and could be incorporated as exogenous macroeconomic covariates in future multivariate studies.

5. Appendix

5.1 Programming Code

In accordance with the project requirements, the complete R code used for data ingestion, exploratory plotting, unit root testing, Box-Tiao transfer function modeling, residual diagnostics, and out-of-sample forecasting is provided below in directly copy-pasteable format:

```r
#
================================================================================
=
#     STSM3744:     Time     Series     Analysis     -     Short     Project
#  Topic:  Interrupted  Time  Series  Analysis  (ITSA)  /  Intervention  Analysis
#  Case  Study:  Impact  of  the  COVID-19  National  Lockdown  (March  2020)  on
#                                  South  African  Consumer  Retail  Trade
#  Data  Source:  Statistics  South  Africa  /  OECD  via  FRED  (Series:
ZAFSLRTTO01IXOBM)
#
================================================================================
=

suppressPackageStartupMessages({
  library(forecast)
  library(tseries)
  library(lmtest)
  library(ggplot2)
  library(TSA)
  library(urca)
})

#         1.         Data         Ingestion         &         Preprocessing
data_file                        <-                        "za_retail_trade.csv"
if                          (!file.exists(data_file))                          {
  url                                                                          <-
"https://fred.stlouisfed.org/graph/fredgraph.csv?id=ZAFSLRTTO01IXOBM"
  download.file(url,     destfile     =     data_file,     mode     =     "wb")
}

raw_data     <-     read.csv(data_file,     stringsAsFactors     =     FALSE)
colnames(raw_data)           <-           c("Date",           "RetailIndex")
raw_data$Date                    <-                    as.Date(raw_data$Date)
raw_data$RetailIndex           <-           as.numeric(raw_data$RetailIndex)
raw_data                         <-                         na.omit(raw_data)

# Estimation Window: Jan 2010 to Dec 2023 (N = 168 Monthly Observations)
data_sub <- raw_data[raw_data$Date >= as.Date("2010-01-01") & raw_data$Date
<=                           as.Date("2023-12-01"),                           ]
data_sub$Month        <-        as.numeric(format(data_sub$Date,        "%m"))
data_sub$Year        <-        as.numeric(format(data_sub$Date,        "%Y"))
data_sub$MonthName <- factor(months(data_sub$Date, abbreviate = TRUE), levels
=                                                                 month.abb)

#  Intervention  Date  (T0):  March  2020;  Reference  Disruption:  July  2021
intervention_date                  <-                  as.Date("2020-03-01")
july2021_date                    <-                    as.Date("2021-07-01")

data_sub$Period  <-  factor(ifelse(data_sub$Date  <  intervention_date,  "Pre-
```

```
Intervention",                                                    "Post-Intervention"),
                          levels      =      c("Pre-Intervention",      "Post-
Intervention"))

retail_ts <- ts(data_sub$RetailIndex, start = c(2010, 1), frequency = 12)
pre_ts <- window(retail_ts, end = c(2020, 2)) # Pre-Intervention: Jan 2010 -
Feb                2020                (n1                 =                 122)

# 2. Figure 1: Time Series Trajectory with March 2020 Intervention & July
2021                                Reference                                  Line
pre_df      <-      data_sub[data_sub$Date      <      intervention_date,      ]
post_df     <-      data_sub[data_sub$Date     >=     intervention_date,     ]

p1    <-    ggplot(data_sub,    aes(x    =    Date,    y    =    RetailIndex))    +
  geom_line(color      =      "#1f77b4",      linewidth      =      0.85)      +
  geom_smooth(data = pre_df, aes(x = Date, y = RetailIndex), method = "lm",
se                                  =                                  FALSE,
              color  =  "#2ca02c",  linetype  =  "dotted",  linewidth  =  0.9)  +
  geom_smooth(data = post_df, aes(x = Date, y = RetailIndex), method = "lm",
se                                  =                                  FALSE,
              color  =  "#ff7f0e",  linetype  =  "dotted",  linewidth  =  0.9)  +
  geom_vline(xintercept = intervention_date, color = "#d62728", linetype =
"dashed",              linewidth              =              1)              +
  geom_vline(xintercept  =  july2021_date,  color  =  "#9467bd",  linetype  =
"dotdash",              linewidth              =              0.9)              +
  annotate("text",    x    =    intervention_date    -    60,    y    =    57,
           label  =  "COVID-19  Intervention\n(March  2020,  T0)",  color  =
"#d62728",    fontface    =    "bold",    size    =    3.3,    hjust    =    1)    +
  annotate("text",    x    =    july2021_date    +    60,    y    =    65,
           label  =  "July  2021  Disruption\n(Civil  Unrest  Ref.)",  color  =
"#9467bd",    fontface    =    "bold",    size    =    3.2,    hjust    =    0)    +
  labs(
    title = "Figure 1: South Africa Monthly Retail Trade Volume Index (2010 -
2023)",
    subtitle = "Showing March 2020 COVID-19 Intervention (Red), July 2021
Reference          (Purple),          and          Regime          Trends",
    x  =  "Date  (Monthly)",  y  =  "Retail  Volume  Index  (2015  =  100)"
  )                                                                          +
  theme_minimal(base_size                          =                          11)
ggsave("figure1_time_series_trajectory.png", plot = p1, width = 9.5, height =
5,                    dpi                    =                    300)

# 3. Descriptive Statistics & Regime Boxplots (Table 1, Figure 2, Figure 3)
skew     <-     function(x)     mean((x     -     mean(x))^3)     /     (sd(x)^3)
kurt     <-     function(x)     mean((x     -     mean(x))^4)     /     (sd(x)^4)
calc_desc                  <-                  function(x)                  {
  c(N  =  length(x),  Mean  =  mean(x),  SD  =  sd(x),  Median  =  median(x),
    IQR = IQR(x), Min = min(x), Max = max(x), Skewness = skew(x), Kurtosis =
kurt(x))
```

```r
}

pre_vals <- data_sub$RetailIndex[data_sub$Period == "Pre-Intervention"]
post_vals <- data_sub$RetailIndex[data_sub$Period == "Post-Intervention"]
desc_table <- data.frame(
  Metric = c("Sample Size (n)", "Mean", "Standard Deviation (SD)", "Median",
             "Interquartile Range (IQR)", "Minimum", "Maximum", "Skewness",
"Kurtosis"),
  Overall = round(calc_desc(data_sub$RetailIndex), 3),
  Pre_Intervention = round(calc_desc(pre_vals), 3),
  Post_Intervention = round(calc_desc(post_vals), 3)
)
print(desc_table)

p2 <- ggplot(data_sub, aes(x = Period, y = RetailIndex, fill = Period)) +
  geom_boxplot(alpha = 0.75, outlier.color = "#d62728", outlier.size = 2,
width = 0.48) +
  scale_fill_manual(values = c("Pre-Intervention" = "#2ca02c", "Post-
Intervention" = "#ff7f0e")) +
  stat_summary(fun = mean, geom = "point", shape = 18, size = 3.5, color =
"black") +
  labs(title = "Figure 2: Pre- vs. Post-Intervention Distribution",
       x = "Regime (T0 = March 2020)", y = "Retail Volume Index (2015 =
100)") +
  theme_minimal(base_size = 11) + theme(legend.position = "none")
ggsave("figure2_pre_post_boxplot.png", plot = p2, width = 5.2, height = 4.5,
dpi = 300)

p3 <- ggplot(data_sub, aes(x = MonthName, y = RetailIndex, fill = Period)) +
  geom_boxplot(alpha = 0.8, outlier.size = 1.3) +
  facet_wrap(~ Period, ncol = 2) +
  scale_fill_manual(values = c("Pre-Intervention" = "#2ca02c", "Post-
Intervention" = "#ff7f0e")) +
  labs(title = "Figure 3: Monthly Seasonal Distribution — Pre-Intervention
vs. Post-Intervention",
       x = "Calendar Month", y = "Retail Volume Index (2015 = 100)") +
  theme_minimal(base_size = 11) + theme(legend.position = "none", axis.text.x
= element_text(angle = 45, hjust = 1))
ggsave("figure3_seasonal_boxplot.png", plot = p3, width = 9.5, height = 4.6,
dpi = 300)

# 4. Stationarity & Baseline Pre-Intervention SARIMA
adf.test(pre_ts)
kpss.test(pre_ts, null = "Level")
d_pre_ts <- diff(pre_ts, differences = 1)
adf.test(d_pre_ts)
kpss.test(d_pre_ts, null = "Level")

base_model <- auto.arima(pre_ts, seasonal = TRUE, stepwise = FALSE,
```

```r
approximation = FALSE)
summary(base_model)

# 5. Intervention Modeling Anchored at T0 = March 2020
t_idx <- 1:nrow(data_sub)
mar2020_idx <- which(data_sub$Date == intervention_date)

pulse_mar2020 <- as.numeric(data_sub$Date == intervention_date)
pulse_apr2020 <- as.numeric(data_sub$Date == as.Date("2020-04-01"))
step_t <- as.numeric(data_sub$Date >= intervention_date)
ramp_t <- pmax(0, t_idx - mar2020_idx) * step_t

xreg_mat1 <- cbind(Pulse_Mar2020 = pulse_mar2020, Lag1_Pulse_Apr2020 =
pulse_apr2020, Step_Mar2020 = step_t)
xreg_mat2 <- cbind(Pulse_Mar2020 = pulse_mar2020, Lag1_Pulse_Apr2020 =
pulse_apr2020, Step_Mar2020 = step_t, Ramp_PostMar2020 = ramp_t)

model_its1 <- Arima(retail_ts, order = c(2, 1, 3), seasonal = list(order =
c(0, 1, 0), period = 12), xreg = xreg_mat1)
model_its2 <- Arima(retail_ts, order = c(2, 1, 3), seasonal = list(order =
c(0, 1, 0), period = 12), xreg = xreg_mat2)
model_its3 <- Arima(retail_ts, order = c(2, 1, 0), seasonal = list(order =
c(0, 1, 1), period = 12), xreg = xreg_mat1)

summary(model_its3)
coeftest(model_its3)

# 6. Residual Diagnostics
resids <- residuals(model_its3)
n_arma_params <- sum(model_its3$arma[1:4])
Box.test(resids, lag = 12, type = "Ljung-Box", fitdf = n_arma_params)
Box.test(resids, lag = 24, type = "Ljung-Box", fitdf = n_arma_params)
Box.test(resids, lag = 36, type = "Ljung-Box", fitdf = n_arma_params)
jarque.bera.test(resids)
shapiro.test(resids)

# 7. Out-of-Sample Forecasting (2023 Holdout Validation & 2024 Ex-Ante
Forecast)
train_ts <- window(retail_ts, end = c(2022, 12))
test_ts <- window(retail_ts, start = c(2023, 1), end = c(2023, 12))
holdout_model <- Arima(train_ts, order = c(2, 1, 0), seasonal = list(order =
c(0, 1, 1), period = 12), xreg = xreg_mat1[1:156, ])
fc_holdout <- forecast(holdout_model, xreg = xreg_mat1[157:168, ], h = 12)
accuracy(fc_holdout, test_ts)

xreg_future_2024 <- cbind(Pulse_Mar2020 = rep(0, 12), Lag1_Pulse_Apr2020 =
rep(0, 12), Step_Mar2020 = rep(1, 12))
fc_2024 <- forecast(model_its3, xreg = xreg_future_2024, h = 12)
print(fc_2024)
```

5.2 Comprehensive AI Prompt Catalog and Critical Reflection

In compliance with the course requirements on the ethical and transparent use of Artificial Intelligence, **Table 7** documents the prompts submitted, the AI responses, and the human verification and corrections applied.

Step #	Project Phase	Prompt Submitted to AI	Summary of AI Output	Human Verification & Statistical Refinement
01	Transfer Function Formulation	*"Provide the Box-Tiao (1975) transfer function equation for an interrupted time series anchored at* $T_0 =$ March 2020 *with an immediate onset pulse* ω_0*, a 1-month lagged closure pulse* $\omega_1 B$*, a step shift, and seasonal ARIMA errors."*	Generated LaTeX equations separating $m_t = (\omega_0 + \omega_1 B)P_t^{(T_0)} + \omega_{\text{step}} S_t^{(T_0)}$ from multiplicative SARIMA noise.	Verified backshift operator algebra against Box, Jenkins, & Reinsel (2015) and aligned notation with STSM3744 course conventions.
02	Data Sourcing & Inspection	*"Identify an authoritative monthly time series dataset measuring South African physical retail volume covering 2010–2024."*	Recommended Stats SA P6242.1 constant 2015=100 volume index via FRED (`ZAFSLRTT01IXOBM`).	Verified all 171 monthly observations (2010–2023 for estimation; Q1 2024 for out-of-sample forecast validation).
03	Visual EDA & Regime Faceting	*"Write R ggplot2 code to plot the retail series with*	Provided `ggplot2` code with `geom_vline()`,	Verified date alignment for March 2020 (T_0) and July

Step #	Project Phase	Prompt Submitted to AI	Summary of AI Output	Human Verification & Statistical Refinement
		vertical lines at March 2020 and July 2021, and create faceted pre- vs post-intervention seasonal boxplots."	`geom_smooth()` regime trends, and `facet_wrap(~ Period)` seasonal boxplots.	2021 reference line; confirmed visual clarity at 300 DPI.
04	Stationarity & ARIMAX Estimation	*"Write R code using `forecast::Arima` and `xreg` to estimate the March 2020 transfer function models and test parameter significance with `lmtest::coeftest`."*	Produced R code for candidate SARIMA and ARIMAX specifications and extracted AIC, BIC, and z-statistics.	Checked degrees of freedom in `Box.test()` (`fitdf = 3` ARMA parameters) and verified white-noise Ljung-Box p-values (> 0.05).
05	Out-of-Sample Forecasting	*"Extend the R script to perform a 12-month holdout test on 2023 and generate a 12-month ahead forecast for 2024 with 80% and 95% confidence ribbons."*	Supplied `window()`, `forecast()`, and `accuracy()` code along with a `geom_ribbon()` forecast plot.	Validated forecasts against actual Q1 2024 Stats SA releases, confirming March 2024 forecast accuracy within 0.01 index points.

Table 7: Structured Catalog of AI Prompts and Human Econometric Validation.

Critical Reflection: Which AI Tools Were Used, and Did AI Reduce Effort or Waste Time?

1. **Specific AI Tools Used:** Throughout this research project, our group utilized **Google Gemini (via the Antigravity AI Coding Assistant)** for R script prototyping, debugging, and document formatting, alongside **OpenAI ChatGPT (GPT-4o)** for initial brainstorming and LaTeX equation structuring.
2. **Did AI Reduce Effort or Was It a Waste of Time?**
    - **Where AI Significantly Reduced Effort and Saved Time:** Overall, using AI was **highly beneficial and substantially reduced our manual workload**, cutting total compilation time from an estimated **24+ hours down to approximately 14 hours** (an effort reduction of $\approx 40\%$). Specifically, AI excelled at automating repetitive coding tasks—such as formatting multi-panel `ggplot2` visualizations, constructing side-by-side Markdown/Word tables, writing boilerplate data-wrangling code, and typesetting complex Box-Tiao transfer function equations in LaTeX. Without AI, debugging `ggplot2` aesthetics and formatting tables would have consumed several hours of trial-and-error.
    - **Where AI Wasted Time and Required Human Correction (Why Human Expertise Was Superior):** Conversely, relying blindly on AI would have caused serious econometric errors that **wasted approximately 2 to 3 hours of our time** before we intervened manually. Most notably, during our initial drafting, the AI assistant naively selected *April 2020* as the sole intervention point simply because April 2020 contained the numerical minimum (53.67), completely ignoring the fact that the South African COVID-19 National State of Disaster and lockdown were enacted in **March 2020**. That initial AI error distorted the post-intervention regime boundary, hid the March 2020 stockpiling surge, and made the post-intervention trajectory look artificially flat rather than increasing. Furthermore, the AI initially miscalculated the degrees of freedom (`fitdf`) in the Ljung-Box test by subtracting exogenous regressors alongside ARMA parameters, and initially plotted overall seasonality without separating pre- and post-intervention regimes.
    - **Overall Verdict:** AI is **not a substitute for statistical thinking**, and in conceptual model specification we could do better—and did do better—through human econometric reasoning and lecturer feedback. However, once we strictly directed the statistical specification ourselves, AI served as a powerful productivity multiplier for R programming and report compilation.